

Calculator Using Postfix Notation

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Doing calculations in a microcontroller-based project is common. Such calculations done by the microcontroller are pre-written as a part of the code, which means that these formulae cannot be changed or altered even if a situation demands so. But what if your system has to solve equations on the fly? What if you want to define an equation to be evaluated and the microcontroller has

to perform the task based on that?

Presented here is a handy postfix notation system that can do your calculations in practically no time, and that too without changing the source code. It can even handle parenthesis and algebraic operator precedence. Instead of explaining just the postfix notation system, algorithm and overall method, let us build a basic calculator based on this technique using Atmel's AVR microcontroller. Fig. 1 shows the author's prototype.



Circuit and working

Fig. 2 shows the circuit diagram of a calculator using postfix notation. The circuit is built around microcontroller ATmega168 (IC1), USB-to-UART converter FT232RL (IC2), 16x2 LCD display (LCD1) and a few easily-available components. You can also use ATmega168 Development Stick, which is available with EFY associates



Fig. 1: Author's prototype

TABLE I
Test Points

Test point	Details
TP0	0V, GND
TP1	5V
TP2	Low when S1 is pressed

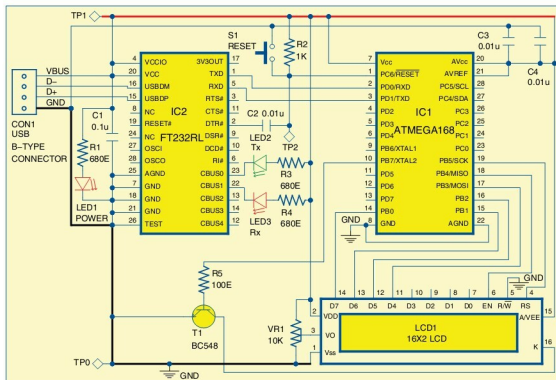


Fig. 2: Circuit diagram of postfix calculator

Kits'n'Spares.

The circuit is powered directly through USB port, which provides 5V supply for the circuit to operate. LED1 indicates presence of power in the circuit. Pins 15 and 16 of IC2 are connected to D+ and D- of USB port and pins 1 and 5 of IC2 are connected to RXD and TXD pins of microcontroller IC1, respectively.

LCD1 is